

Next-generation Bayesian local robustness

Ryan Giordano (rgiordano@berkeley.edu, UC Berkeley)

Berkeley Neyman Seminar

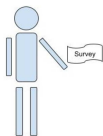
September 2026

The basic problem

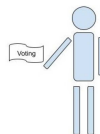
We have a survey population, for whom we observe:

- Covariates x (e.g. race, gender, zip code, age, education level)
- Responses y (e.g. A binary response to “do you support candidate Z”)

We want the average response in a target population, in which we observe only covariates.



Observe (x_i, y_i) for $i = 1, \dots, N_S$
(This is the “Data”, $\mathcal{D}$)



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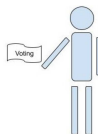
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Photo copyright Mark Taylor: natereport.com

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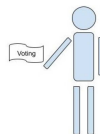
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This talk will study “Multilevel regression and poststratification” estimators:

- Set $\mathbb{E}[y|x, \theta] = m(\theta^\top x)$ and prior $\mathcal{P}(\theta)$ (e.g. Hierarchical logistic regression)
- Estimate the posterior $\mathcal{P}(\theta|\mathcal{D})$ with Markov Chain Monte Carlo (MCMC)
- Take $\hat{\mu} = \frac{1}{N_T} \sum_{j=1}^{N_T} \hat{y}_j$ where $\hat{y}_j = \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}[y|x_j]$.

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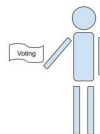
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Problem: MCMC is time consuming and the regression model is surely misspecified.

Goal: Meaningful regression specification checks with frequentist uncertainty.

Result preview

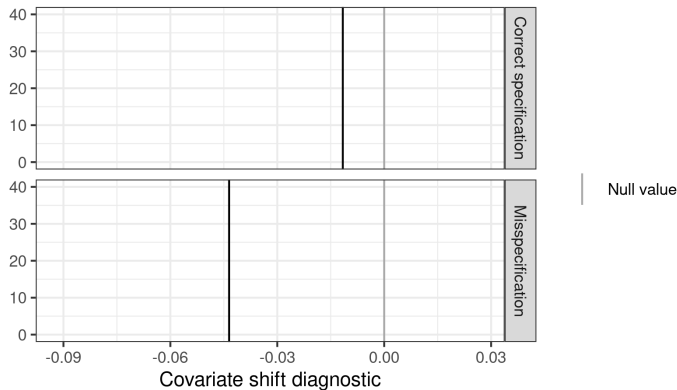


Figure 1: A covariate shift diagnostic

Here is our check for two models on the same simulated data: one correctly specified and one misspecified. The misspecified model is more extreme.

Problem: What is “large enough” to be real evidence of misspecification?

Possible answer: Frequentist confidence intervals via the parametric bootstrap.

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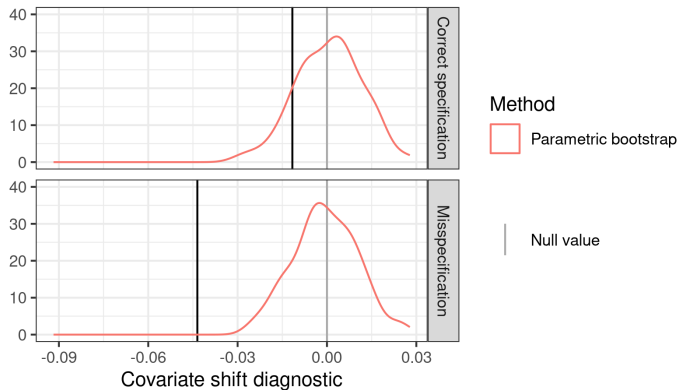


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The frequentist interval based on the parametric bootstrap rejects the misspecified model but not the correctly specified model.

Problem: Expensive: each bootstrap sample required a new MCMC run.

Possible answer: Infinitesimal jackknife confidence intervals? [GB26]

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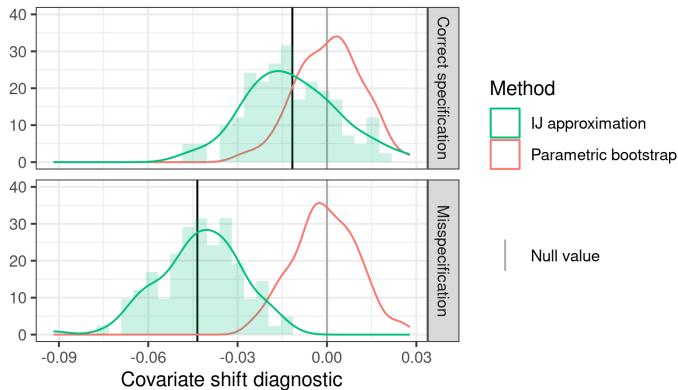


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IJ confidence intervals are fast and often work well for posterior means.
But they are not accurate for this more complex diagnostic (high Monte Carlo error).

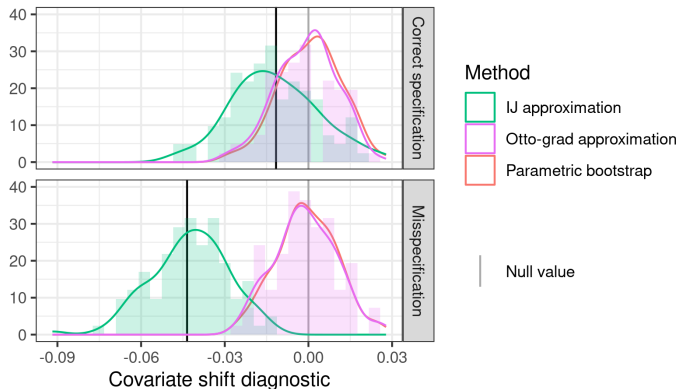
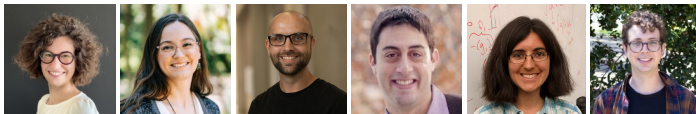


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Answer: Local glow-based sensitivity measures produce good approximations to the bootstrap at negligible computational expense.

- Local checks of specification of Bayesian regression models
 - (With Alice Cima, Erin Hartman, Jared Miller, Avi Feller)
- Frequentist variability of Bayesian posteriors with the infinitesimal jackknife
 - (With Tamara Broderick)
- Bayesian robustness with local flows
 - (With Lucas Schwengber)



Local checks of specification of Bayesian regression models

$$\hat{\mu} = \frac{1}{N_T} \sum_{j=1}^{N_T} \hat{y}_j = \underbrace{\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} \left[\frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\top x_j) \right]}_{\text{Want to check specification without re-running MCMC}}.$$

Some existing local approaches:

- Specify an alternative model, maybe test with Bayes factor [DRM25; CBR25]
 - Bayes factors can depend strongly on priors (Lindley's paradox)
 - Articulating a family of models is inconvenient
 - Doesn't take the target into account
- Use out-of-sample checking [VGG17; Kuh+24]
 - The task is not to predict on a new IID survey datapoint

Local specification checks

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The model is definitely misspecified, we want to check whether it's *importantly* misspecified.

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Let $\mathcal{P}(x, y)$ denote a generic joint density of x, y .

Let $\hat{\mathcal{P}}_S$ the empirical distribution on the survey data.

1. Write our estimand as a functional $\mu(\mathcal{P})$ that takes in a density and returns a MrP estimate
2. Define a covariate density perturbation $w(x)$
3. Define $\mathcal{P}_\epsilon(x, y) = \mathcal{P}(x, y) + \epsilon w(x) \mathcal{P}(y|x) = (\mathcal{P}(x) + \epsilon w(x)) \mathcal{P}(y|x)$
4. Compute the derivative $\Delta(w, \mathcal{P}) = \partial \mu(\mathcal{P}_\epsilon) / \partial \epsilon$
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Lots of similar ideas in causal inference, kernel learning, transfer learning [Shi00; Böh20; Roj+; MJK20]. This is just a local Bayesian diagnostics version.

Simple visual example

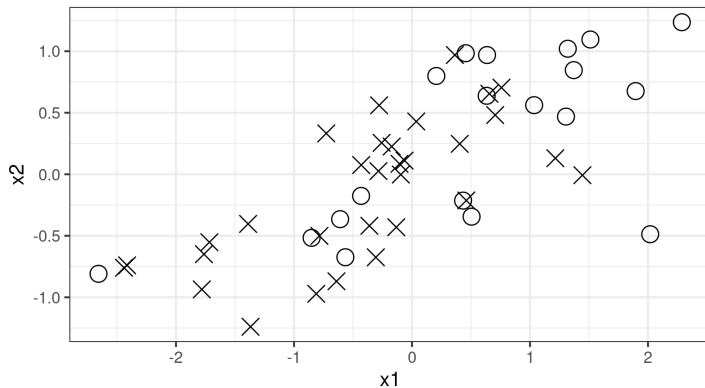


Figure 2: A covariate shift diagnostic

Consider logistic regression with two regressors, x_1 and x_2 .

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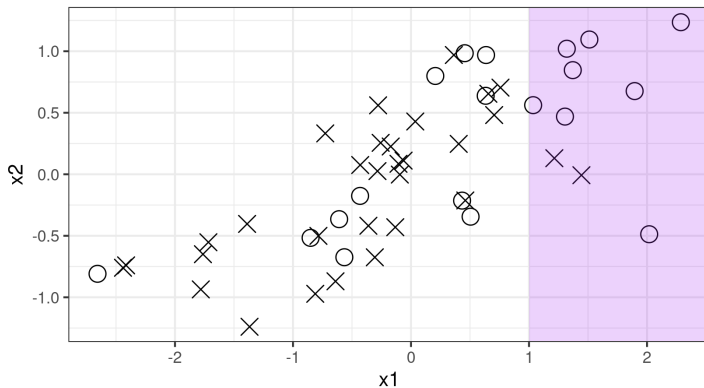


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Taking $w(x) = \mathbb{I}(x_1 > 1)$, we upweight positive examples.

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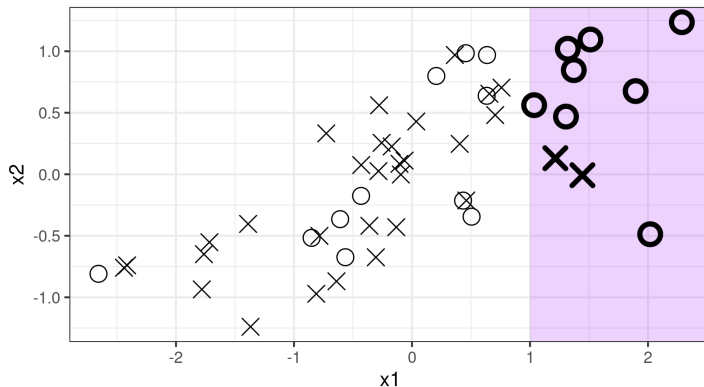


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With enough data, a regression that omits x_1 will change its coefficient of x_2 under re-weighting, whereas a correctly specified regression will not need to.

Let the regression log likelihood be $\ell(y|x, \theta)$, and fix $w(x)$. Then define

$$\mathcal{P}(\theta|\mathcal{D}) \propto \exp\left(\sum_{i=1}^{N_S} \ell(y_i|x_i, \theta)\right) \pi(\theta) \text{ and } \mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta|\mathcal{D}) \exp\left(\sum_{i=1}^{N_S} \epsilon w(x_i) \ell(y_i|x_i, \theta)\right)$$

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Theorem: [DF86; Gus96; GBJ18]:

Consider a family of distributions $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$ and a function $\phi(\theta)$. By interchange of integration and differentiation,

$$\left. \frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)]}{\partial \epsilon} \right|_{\epsilon=0} = \text{Cov}_{\mathcal{P}(\theta)} (\phi(\theta), h(\theta)) .$$

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Our diagnostic is then given by

$$\Delta(\mathcal{D}) := \left. \frac{\partial \hat{\mu}}{\partial \epsilon} \right|_{\epsilon=0} = \text{Cov}_{\mathcal{P}(\theta|\mathcal{D})} \left(\underbrace{\frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\top x_j)}_{\text{Estimand}}, \underbrace{\sum_{i=1}^{N_S} w(x_i) \ell(y_i|x_i, \theta)}_{\text{Log model perturbation}} \right)$$

We expect $\Delta(w, \mathcal{D})$ to be *close* to zero when the model is correctly specified.

Bootstrapping Bayesian sensitivity analysis

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But on any actual dataset, $\Delta(w, \mathcal{D})$ will typically not be precisely zero, even under correct specification.

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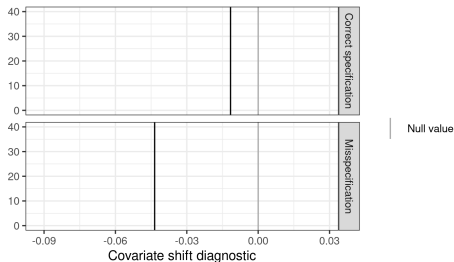


Figure 3: A covariate shift diagnostic

Simulation data: Logistic regression with five active regressors, each with covariate shift.

Model 0 (correct) : $y \sim \text{Logistic}(1 + X1 + X2 + X3 + X4 + X5)$

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To assess “closeness to zero,” compare with $\Delta(\mathcal{D}^*)$, where $\mathcal{D}^*$ is a bootstrap sample.

- Compute $\Delta(\mathcal{D})$ on the original dataset
- For $b = 1, \dots, 100$:
 1. Sample $\mathcal{D}_b^* := (x_1^*, y_1^*), \dots (x_{N_S}^*, y_{N_S}^*)$ (parametric or nonparametric bootstrap)
 2. Run MCMC to estimate $\mathcal{P}(\theta|\mathcal{D}_b^*)$ (**time consuming!**)
 3. Use the draws to estimate $\Delta(\mathcal{D}_b^*)$
- Compare Δ with distribution of $\Delta(\mathcal{D}_b^*)$.

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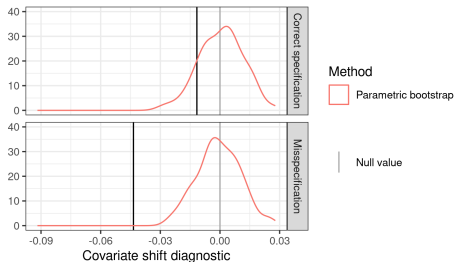


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Frequentist variance with the infinitesimal
jackknife

Why frequentist variance?

For the remainder of the talk, focus on *fast approximations* to the *frequentist* sampling distribution under $x_i, y_i \stackrel{iid}{\sim} \mathcal{P}_S(x, y)$ of Bayesian posterior quantities like our diagnostic.

We will form *local approximations* computable only from *the original posterior*.

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No need to commit to our covariate shift diagnostic to care!

- Frequentist variability of *any* model checking diagnostic is interesting
- Frequentist variability remains meaningful under model misspecification [HM23; WDH26; LJ26; JLR25]

We will compare with computationally expensive full bootstrap.

Data re-weighting (again)

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If this series is accurate, then we can compute ψ_i using the original posterior and quickly approximate bootstrap draws.

Bayesian von-Mises Expansion

Define the “generalized posterior” functional [GB26]:

$$T(\mathcal{P}, N) := \frac{\int \phi(\theta) \exp \left(N \int \ell(y|x, \theta) \mathcal{P}(x, y) dx dy \right) \pi(\theta) d\theta}{\int \exp \left(N \int \ell(y|x, \theta) \mathcal{P}(x, y) dx dy \right) \pi(\theta) d\theta}.$$

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To derive frequentist variance estimates, we study the *von Mises expansion* [von47].

Let $\tilde{\mathcal{P}}^t = t\hat{\mathcal{P}}_S + (1-t)\mathcal{P}_S$. Then form the series expansion in t :

$$\begin{aligned} \sqrt{N_S} \left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] - T(\mathcal{P}_S, N_S) \right) &= \sqrt{N_S} \left(T(\hat{\mathcal{P}}_S, N_S) - T(\mathcal{P}_S, N_S) \right) \\ &= \sqrt{N_S} \left. \frac{\partial T(\tilde{\mathcal{P}}^t, N_S)}{\partial t} \right|_{t=0} (1-0) + \mathcal{E}(\tilde{t}) \quad (\tilde{t} \in [0, 1]) \\ &= \sqrt{N_S} \underbrace{\sum_{n=1}^N (\psi_i - \bar{\psi})}_{\text{Infinitesimal jackknife estimator}} + o_p(1) + \mathcal{E}(\tilde{t}). \end{aligned}$$

Bayesian von-Mises Expansion Results

$$\sqrt{N_S} \left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] - T(\mathcal{P}_S, N_S) \right) = \sqrt{N_S} \sum_{n=1}^N (\psi_i - \bar{\psi}) + o_p(1) + \mathcal{E}(\tilde{t}).$$

Giordano and Broderick [GB26, Theorem 3] (sketch):

Under Bernstein-von Mises-like conditions, $\sup_{\tilde{t} \in [0,1]} |\mathcal{E}(\tilde{t})| \xrightarrow{P} 0$.

Corollary [von47]: Let $\frac{1}{N_S} \sum_{i=1}^{N_S} (\psi_i - \bar{\psi})^2 \xrightarrow{P} V$. By the CLT applied to ψ_i ,

$$\sqrt{N_S} \left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] - T(\mathcal{P}_S, N_S) \right) \xrightarrow{d} \mathcal{N}(0, V).$$

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To compute and study the parametric bootstrap instead is a simple change:

$$\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta|\mathcal{D}) \exp \left(\sum_{i=1}^{N_S} \epsilon (\ell(y_i^*|x_i, \theta) - \ell(y_i|x_i, \theta)) \right).$$

(See, e.g., Giordano et al. [Gio+26] Theorem 4.1).

Bayesian von-Mises Expansion Results

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Note: The von-Mises expansion is stronger than necessary for consistency [GB26]. But it can be used to prove *lower bounds* on $\mathcal{E}(\tilde{t})$ and so *inconsistency* when no BVM theorem holds.

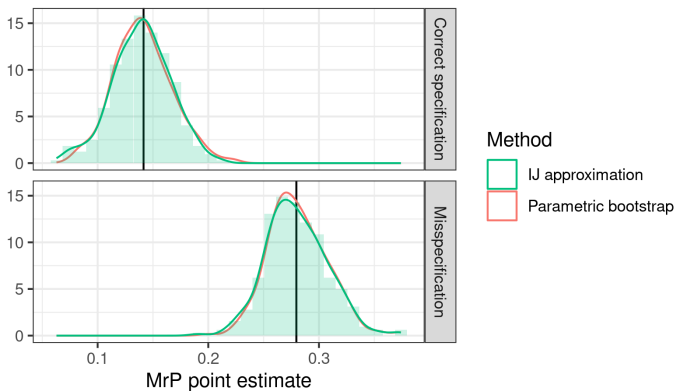


Figure 4: MrP Posterior Mean

The IJ does great for the MrP point estimate. See also Luo and Ji [LJ26] and Ji, Lee, and Rabe-Hesketh [JLR25].

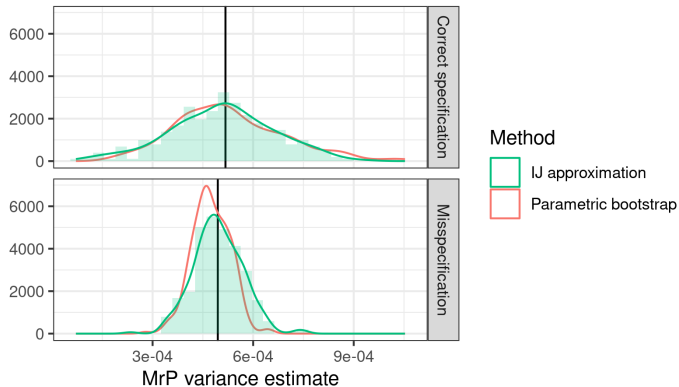


Figure 4: MrP Posterior Variance

The IJ does pretty well for the MrP posterior variance.

Data re-weighting.

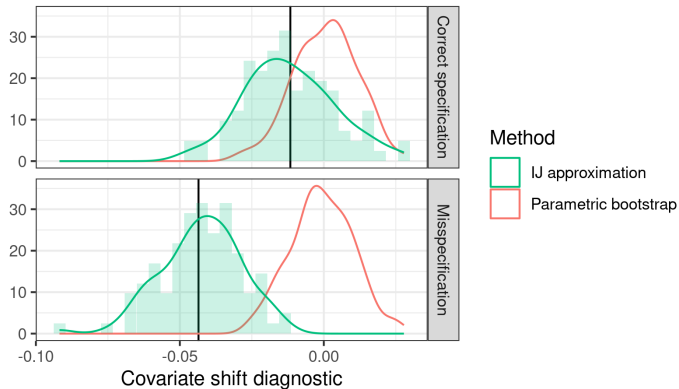


Figure 4: Covariate Shift Diagnostic

Need something better than the IJ for noisier / more complex posterior quantities!

Bayesian robustness with local flows

Why flows for sensitivity analysis?

Current Bayesian local robustness is built around linearizing posterior expectations:

$$\mathbb{E}_{\mathcal{P}^{\epsilon}(\theta)} [\phi(\theta)] \approx \mathbb{E}_{\mathcal{P}(\theta)} [\phi(\theta)] + \left. \frac{\partial \mathbb{E}_{\mathcal{P}^{\epsilon}(\theta)} [\phi(\theta)]}{\partial \epsilon} \right|_{\epsilon=0} (\epsilon - 0),$$

where the derivative is estimated — with noise — from MCMC draws [Gus96; GBJ18; CBR25; DRM25; Gio+26].

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This has some shortcomings:

- Potential MCMC variance, not obvious how to smooth
- Linear extrapolation can lead to impossible values (e.g. negative variances)
- Complex quantities like posterior quantiles are hard to linearize
- Bayesians like to work with draws, not analytic expectations [HGV25]

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Idea: Instead of linearizing expectations, learn a flow from $\mathcal{P}$ to $\mathcal{P}^{\epsilon}$.

There's a large literature on learning flows between distant distributions. [Pap+21]

Our problem is easier:

- Only need “local” flows
- Our perturbations have a lot of structure (e.g. bootstraps)

Consider $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$, with draws $\theta \sim \mathcal{P}(\theta)$.

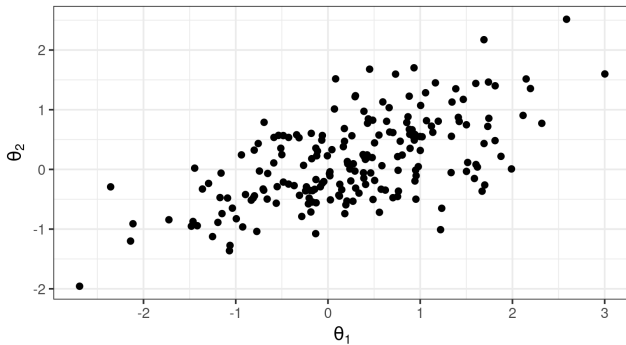


Figure 5: Draws from a posterior

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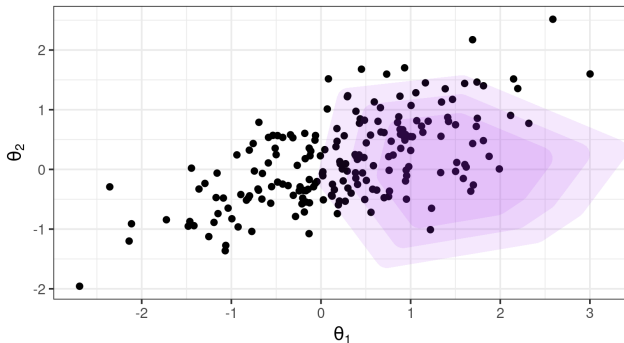


Figure 5: A posterior perturbation

Imagine the density $\mathcal{P}(\theta)$ coming out of the page. The perturbation $h(\theta)$ is defined “vertically.”

Consider $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$, with draws $\theta \sim \mathcal{P}(\theta)$.

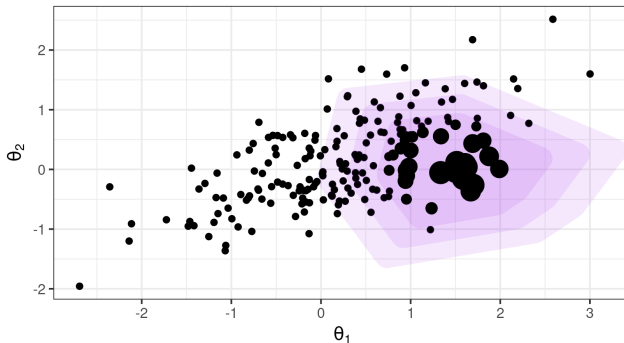


Figure 5: A posterior perturbation with weights

Plug-in estimates for the covariance are equivalent to producing the perturbation by weighting [GBJ18, Appendix B]

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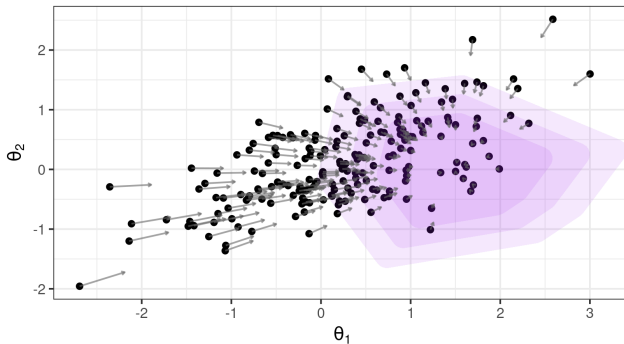


Figure 5: A posterior perturbation with a flow

We will try, instead, try to effect the perturbation with a flow $\theta \mapsto \Phi^\epsilon(\theta)$

Consider $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$.

We want a flow $\theta \mapsto \Phi^\epsilon(\theta)$ such that, for small ϵ , $\Phi^\epsilon(\theta) \sim \mathcal{P}^\epsilon(\theta)$ and $\Phi^0(\theta) = \theta \sim \mathcal{P}(\theta)$.

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Suppose there is a scalar-valued potential function $\phi(\theta)$ such that

$$\Phi^\epsilon(\theta) = \theta + \epsilon \nabla \phi(\theta) + O(\epsilon^2) \quad \text{as } \epsilon \rightarrow 0. \text{ Then:}$$

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Theorem: [Joh00; KV86; BGL14], Schwengber 2026:

Assume that $\mathbb{E}_{\mathcal{P}(\theta)} [h(\theta)^2] < \infty$ and $\mathcal{P}(\theta)$ has sub-Gaussian tails. Then there exists a solution ϕ to the differential equation

$$\mathcal{L}_{\mathcal{P}}(\phi) := -\nabla \phi^\top \nabla \log \mathcal{P} - \Delta \phi = h$$

such that $\|\nabla \phi\|_{L^2(\mathcal{P})} < \infty$ and the flow $\theta_\epsilon = \theta + \epsilon \nabla \phi(\theta)$ satisfies Equation $(\star)$.

Note that $\mathcal{L}_{\mathcal{P}}$ is well-known: it's the drift operator, generator of a Langevin diffusion, and a Stein operator.[ABS+21; Son+21; Ros11]

How can we find $\phi(\theta)$ in practice?

Solving PDEs is hard. How can we easily find $\phi(\theta)$ in practice?

$$(\star) \quad \text{Cov}_{\mathcal{P}(\theta)}(\phi(\theta), h(\theta)) = \mathbb{E}_{\mathcal{P}(\theta)}[\nabla \phi(\theta)^\top \nabla \phi(\theta)] \quad (\text{Identity connecting } h \text{ and } \phi)$$

Basis expansion approach (the Galerkin method [EG04]):

- Specify a family of real-valued target functions $\phi_n(\theta)$ for $n \in [N]$
- Specify a family of real-valued basis functions $b_m(\theta)$ for $m \in [M]$

Let $\phi(\theta; \beta) := \sum_{m=1}^M \beta_m b_m(\theta)$ and plug into Equation $(\star)$.

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Stacking in matrices:

$$\underbrace{\begin{bmatrix} \vdots \\ \text{Cov}_{\mathcal{P}(\theta)}(\phi_n(\theta), h(\theta)) \\ \vdots \end{bmatrix}}_{N \times 1 \text{ matrix}} \underbrace{\begin{bmatrix} \vdots \\ \dots \quad \mathbb{E}_{\mathcal{P}(\theta)}[\nabla \phi_n(\theta)^\top \nabla b_m(\theta)] \quad \dots \\ \vdots \end{bmatrix}}_{N \times M \text{ matrix}} = \begin{bmatrix} \vdots \\ \beta_m \\ \vdots \end{bmatrix}$$

This gives N linear equations in M unknowns, and the matrices can be estimated with MCMC.

Theorem [GK69; BGL14], Schwengber 2026: When $\mathcal{P}(\theta)$ is Gaussian, taking $N = M$ and $\phi_n(\theta) = b_n(\theta)$ to be the Hermite polynomials give the best finite-rank approximation to $\mathcal{L}_{\mathcal{P}}^{-1}$ in its operator norm.

Using the 1^{st} order Hermite polynomials $\phi_n(\theta) = b_n(\theta) = \theta_n$, the estimated flow is given by

$$\theta_{\epsilon} = \theta + \epsilon \underbrace{\text{Cov}_{\mathcal{P}(\theta|\mathcal{D})}(\theta, h(\theta))}_{\left. \frac{\partial \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}[\theta]}{\partial \epsilon} \right|_{\epsilon=0}}$$

This just moves all points in the direction of the mean shift!

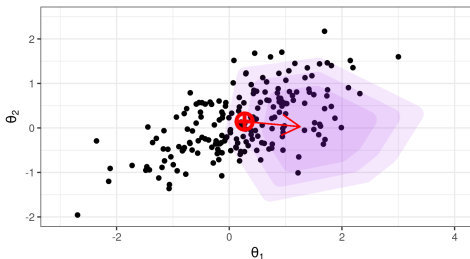
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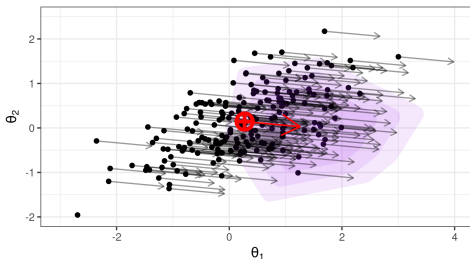
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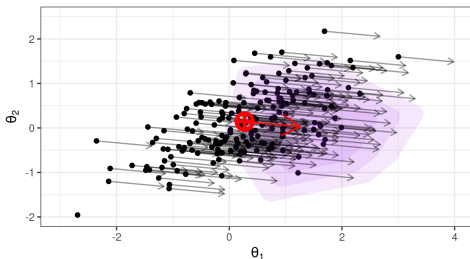
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Suppose we have $\theta_m \sim \mathcal{P}(\theta|\mathcal{D})$, and bootstrap weights w_i .

Set $\theta_m^* = \theta_m + \frac{1}{N_S} \sum_{i=1}^{N_S} w_i \psi_i$ and compute any posterior quantities from these samples.

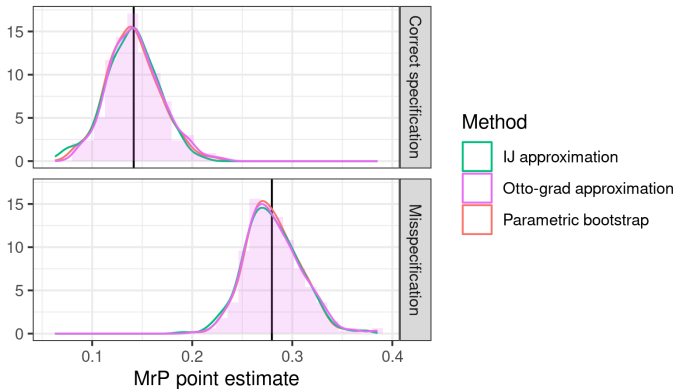


Figure 6: MrP Posterior Mean

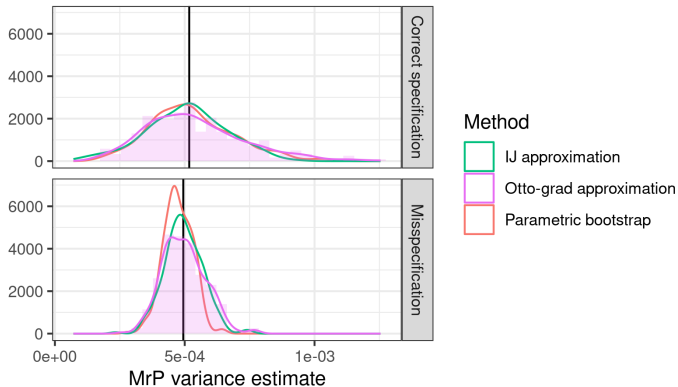


Figure 6: MrP Posterior Variance

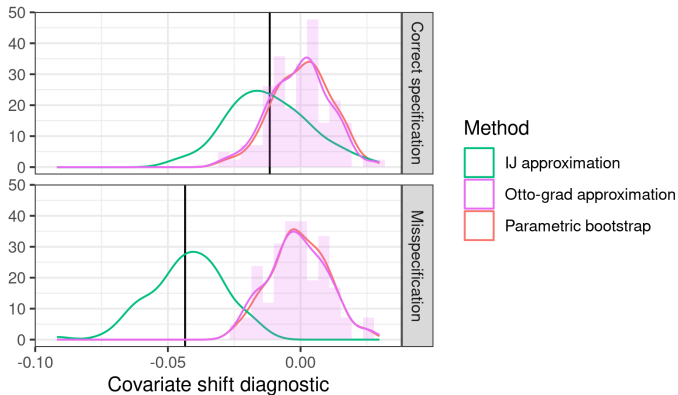


Figure 6: Covariate Shift Diagnostic

Carefully understanding the gap between IJ and the flow is ongoing work!

- Local checks of specification of Bayesian regression models
 - Use invariance to covariate shift as a diagnostic for Bayesian hierarchical models
- Frequentist variability of Bayesian posteriors with the infinitesimal jackknife
 - Use sensitivity of mean parameters to speed up bootstrapping MCMC
- Bayesian robustness with local flows
 - Use easy-to-compute approximate flows to speed up bootstrapping MCMC

- [ABS+21] Luigi Ambrosio, Elia Brué, Daniele Semola, et al. *Lectures on optimal transport*. Vol. 130. Springer Cham, 2021.
- [BGL14] Dominique Bakry, Ivan Gentil, and Michel Ledoux. *Analysis and geometry of Markov diffusion operators*. Vol. 103. Springer, 2014.
- [Büh20] Peter Bühlmann. “Invariance, Causality and Robustness: 2018 Neyman Lecture”. In: *Statistical Science* 35.3 (2020), pp. 404–426. ISSN: 0883-4237. JSTOR: 26997912. URL: <https://www.jstor.org/stable/26997912> (visited on 08/18/2026).
- [CBR25] Rafael Cabral, David Bolin, and Håvard Rue. “Robustness, Model Checking, and Hierarchical Models”. In: *Journal of the Royal Statistical Society Series B: Statistical Methodology* 87.3 (July 14, 2025), pp. 632–652. ISSN: 1369-7412, 1467-9868. DOI: 10.1093/jrsssb/qkae107. URL: <https://academic.oup.com/jrsssb/article/87/3/632/7914952> (visited on 08/31/2026).
- [DF86] P. Diaconis and D. Freedman. “On the Consistency of Bayes Estimates”. In: *The Annals of Statistics* (1986), pp. 1–26.

- [DRM25] Antonio Di Noia, Fabrizio Ruggeri, and Antonietta Mira. “Likelihood Distortion and Bayesian Local Robustness”. In: *Bayesian Analysis* 20.4 (Dec. 1, 2025). ISSN: 1936-0975. DOI: 10.1214/25-BA1526. URL: <https://projecteuclid.org/journals/bayesian-analysis/volume-20/issue-4/Likelihood-Distortion-and-Bayesian-Local-Robustness/10.1214/25-BA1526.full> (visited on 09/01/2026).
- [EG04] Alexandre Ern and Jean-Luc Guermond. *Theory and Practice of Finite Elements*. en. Ed. by S. S. Antman, J. E. Marsden, and L. Sirovich. Vol. 159. Applied Mathematical Sciences. New York, NY: Springer, 2004. ISBN: 978-1-4419-1918-2 978-1-4757-4355-5. DOI: 10.1007/978-1-4757-4355-5. URL: <http://link.springer.com/10.1007/978-1-4757-4355-5> (visited on 01/06/2026).
- [GB26] **R. Giordano and T. Broderick.** “The Bayesian Infinitesimal Jackknife for Variance”. In: *arXiv preprint arXiv:2305.06466* (2026).
- [GBJ18] **R. Giordano, T. Broderick, and M. Jordan.** “Covariances, robustness and variational Bayes”. In: *The Journal of Machine Learning Research* 19.1 (2018), pp. 1981–2029.

- [Gio+26] R. Giordano et al. *Locally Equivalent Weights for Multilevel Regression and Poststratification*. 2026. arXiv: 2606.04250 [stat.ME]. URL: <https://arxiv.org/abs/2606.04250>.
- [GK69] Israel Gohberg and M Kreĭn. “Introduction to the theory of linear nonselfadjoint operators in Hilbert space”. In: *Translations of mathematical monographs* (1969).
- [Gus96] P. Gustafson. “Local sensitivity of posterior expectations”. In: *The Annals of Statistics* 24.1 (1996), pp. 174–195.
- [HGV25] Geonhee Han, Andrew Gelman, and Aki Vehtari. *Efficient Scenario Analysis in Real-Time Bayesian Election Forecasting via Sequential Meta-Posterior Sampling*. arXiv.org. Oct. 21, 2025. URL: <https://arxiv.org/abs/2510.19133v1> (visited on 05/27/2026).
- [HM23] J. Huggins and J. Miller. “Reproducible model selection using bagged posteriors”. In: *Bayesian Analysis* 18.1 (2023), pp. 79–104.

- [JLR25] Feng Ji, JoonHo Lee, and Sophia Rabe-Hesketh. “Valid Standard Errors for Bayesian Quantile Regression With Clustered and Independent Data”. In: *Journal of Educational and Behavioral Statistics* (2025), p. 10769986251379738. ISSN: 1076-9986. DOI: 10.3102/10769986251379738. URL: <https://doi.org/10.3102/10769986251379738> (visited on 08/31/2026).
- [Joh00] Jon Johnsen. “On the Spectral Properties of Witten-Laplacians, Their Range Projections and Brascamp-Lieb’s Inequality”. In: *Integral Equations and Operator Theory* 36.3 (Sept. 2000), pp. 288–324. ISSN: 0378-620X, 1420-8989. DOI: 10.1007/BF01213926. arXiv: 1703.10011 [math-ph]. URL: <http://arxiv.org/abs/1703.10011> (visited on 09/03/2026).
- [Kuh+24] Swen Kuh et al. “Using Leave-One-out Cross Validation (LOO) in a Multilevel Regression and Poststratification (MRP) Workflow: A Cautionary Tale”. In: *Statistics in Medicine* 43.5 (2024), pp. 953–982. ISSN: 1097-0258. DOI: 10.1002/sim.9964. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/sim.9964> (visited on 05/18/2026).

- [KV86] C. Kipnis and S. R. S. Varadhan. “Central Limit Theorem for Additive Functionals of Reversible Markov Processes and Applications to Simple Exclusions”. In: *Communications in Mathematical Physics* 104.1 (Mar. 1986), pp. 1–19. ISSN: 0010-3616, 1432-0916. DOI: 10.1007/BF01210789. URL: <http://link.springer.com/10.1007/BF01210789> (visited on 09/03/2026).
- [LJ26] Nanyu Luo and Feng Ji. *Robust Standard Errors for Bayesian Posterior Functionals via the Infinitesimal Jackknife*. 2026. DOI: 10.48550/arXiv.2604.03398. arXiv: 2604.03398 [stat.ME]. URL: <http://arxiv.org/abs/2604.03398> (visited on 07/24/2026). Pre-published.
- [MJK20] Krikamol Muandet, Wittawat Jitkrittum, and Jonas Kübler. “Kernel Conditional Moment Test via Maximum Moment Restriction”. In: *Proceedings of the 36th Conference on Uncertainty in Artificial Intelligence (UAI)*. Conference on Uncertainty in Artificial Intelligence. PMLR, Aug. 27, 2020, pp. 41–50. URL: <https://proceedings.mlr.press/v124/muandet20a.html> (visited on 08/31/2026).

- [Pap+21] G. Papamakarios et al. “Normalizing flows for probabilistic modeling and inference”. In: *Journal of Machine Learning Research* 22:57 (2021), pp. 1–64.
- [Roj+] Mateo Rojas-Carulla et al. “Invariant Models for Causal Transfer Learning”. In: ().
- [Ros11] Nathan Ross. “Fundamentals of Stein’s method”. In: (2011).
- [Shi00] Hidetoshi Shimodaira. “Improving Predictive Inference under Covariate Shift by Weighting the Log-Likelihood Function”. In: *Journal of Statistical Planning and Inference* 90.2 (Oct. 2000), pp. 227–244. ISSN: 03783758. DOI: 10.1016/S0378-3758(00)00115-4. URL: <https://linkinghub.elsevier.com/retrieve/pii/S0378375800001154> (visited on 09/03/2026).
- [Son+21] Yang Song et al. *Score-Based Generative Modeling through Stochastic Differential Equations*. 2021. arXiv: 2011.13456 [cs.LG]. URL: <https://arxiv.org/abs/2011.13456>.

- [VGG17] Aki Vehtari, Andrew Gelman, and Jonah Gabry. “Practical Bayesian Model Evaluation Using Leave-One-out Cross-Validation and WAIC”. In: *Statistics and Computing* 27.5 (Sept. 1, 2017), pp. 1413–1432. ISSN: 1573-1375. DOI: 10.1007/s11222-016-9696-4. URL: <https://doi.org/10.1007/s11222-016-9696-4> (visited on 05/27/2026).
- [von47] R. von Mises. “On the asymptotic distribution of differentiable statistical functions”. In: *The Annals of Mathematical Statistics* 18.3 (1947), pp. 309–348.
- [WDH26] Yu Wang, Jie Ding, and Jonathan H. Huggins. *Accurate Large-sample Uncertainty Quantification Using Stochastic Gradient Markov Chain Monte Carlo*. 2026. DOI: 10.48550/arXiv.2606.00293. arXiv: 2606.00293 [cs.LG]. URL: <http://arxiv.org/abs/2606.00293> (visited on 08/31/2026). Pre-published.